Deepak Sathyan

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Mitchell Institute for Fundamental Physics and Astronomy, Texas A&M University, College Station, TX

Education

University of Maryland — College Park, MD
Ph.D. in Physics — Aug 2018 – July 2024
Thesis: Unifying Searches for New Physics with Precision Measurements of the W Boson Mass
Advisor: Dr. Kaustubh Agashe

Boston University — Boston, MA
Bachelor of Arts in Physics, Summa Cum Laude with Honors — Aug 2014 – May 2018
GPA: 3.9
Minor: Mathematics
Thesis: Muon (g-2): Searching for the Muon Electric Dipole Moment
Advisor: Dr. Robert Carey

Experience

Postdoctoral Research Associate — College Station, TX
Mitchell Institute for Fundamental Physics and Astronomy — Sept 2024 – Aug 2027
Texas A&M University — 3 years

Research Interests

Beyond the Standard Model searches with Collider Physics
searches at the LHC, sub-GeV dark matter, invisible new physics via precision measurements, Kaluza-Klein modes in Randall-Sundrum models, baryon number violation models, muon collider phenomenology

Searches for sub-GeV Dark Matter
Constraints on dark matter-neutrino interactions, dark photon production at the LHC, dark sector production at muon collider, astrophysical signals of cosmic ray-dark matter scattering, astrophysical sources of sub-GeV dark matter

Neutrino Physics
Supernova neutrino attenuation and detection, neutrino oscillations in open systems, neutrinophilic mediators

Articles in Review

3. **Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering** — May 2026
B. Dutta, D. Goswami, J. Kumar, M. Rai, D. Sathyan
arxiv.org/abs/2605.08010

2. **A Baryon and Lepton Number Violation Model Testable at the LHC** Aug 2025
   A. Bhoonah, F. Burk, D. Liu, T. Ou, D. Sathyan
   arxiv.org/abs/2508.21064

1. **New Constraints on Neutrino-Dark Matter Interactions: A Comprehensive Analysis** July 2025
   P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
   arxiv.org/abs/2507.01000

Articles in Refereed Journals

5. **New laboratory constraints on neutrinophilic mediators** July 2024
   P.S.B. Dev, D. Kim, D. Sathyan, K. Sinha, Y. Zhang
   arxiv.org/abs/2407.12738 (Phys.Lett.B 868 (2025) 139765)

4. **'Unification' of BSM searches and SM measurements: the case of lepton+MET and m_W** Apr 2024
   K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
   arxiv.org/abs/2404.17574 (JHEP 02 (2025) 139)

3. **A new purpose for the W-boson mass measurement: Searching for New Physics in lepton+MET** Oct 2023
   K. Agashe, S. Airen, R. Franceschini, D. Kim, A.V. Kotwal, L. Ricci, D. Sathyan
   arxiv.org/abs/2310.13687 (Phys.Lett.B 855 (2024) 138774)

2. **Energy-peak based method to measure top quark mass via B-hadron decay lengths** Dec 2022
   K. Agashe, S. Airen, R. Franceschini, J. Incandela, D. Kim, D. Sathyan
   arxiv.org/abs/2212.03929 (JHEP 06 (2023) 021)

1. **LHC Signals for KK Graviton from an Extended Warped Extra Dimension** Aug 2020
   K. Agashe, M. Ekhterachian, D. Kim, D. Sathyan
   arxiv.org/abs/2008.06480 (JHEP 11 (2020) 109)

Articles in Preparation

1. **LHC as a Beam Dump: Probing Light Dark Photons** Sept 2026
   B. Dutta, A. Karthikeyan, D. Kim, H. Kim, T. Kim, D. Sathyan
   arxiv.org/a/sathyan_d_1

2. **New Probes of Dark Sectors at a Muon Collider** Jan 2027
   B. Dutta, A. Karthikeyan, D. Kim, K. Kelly, D. Sathyan
   arxiv.org/a/sathyan_d_1

3. **Exploring open systems approach to neutrino oscillations** Mar 2027
   A. Chandra Shekar, K. Kelly, D. Sathyan, L. Strigari, T. Zhou
   arxiv.org/a/sathyan_d_1

4. **LHC Signals for KK Gravitons Producing 8-Particle Final States** Mar 2027
   K. Agashe, D. Kim, P. Maksimovic, S. Mondal, M. Osherson, K. Panchal, D. Plotnikov, D. Sathyan
   arxiv.org/a/sathyan_d_1

Ph.D. Thesis

1. **Unifying Searches for New Physics with Precision Measurements of the W Boson Mass** 2024
   D. Sathyan
   drum.lib.umd.edu/items/f9b081c8-444a-4af2-b69e-c8156d11a2c9 (Ph.D. Thesis)

Snowmass2021 Contributions

5. **TF07 Snowmass Report: Theory of Collider Phenomena** Oct 2022
   F. Maltoni et al.
   arxiv.org/abs/2210.02591

4. **Report of the Topical Group on Top quark physics and heavy flavor production for Snowmass 2021** Sept 2022
   K. Agashe et al.
   arxiv.org/abs/2209.11267

3. **Report of the Topical Group on Physics Beyond the Standard Model at Energy Frontier for Snowmass 2021** Sept 2022
   T. Bose et al.
   arxiv.org/abs/2209.13128

2. **Snowmass2021 - White Paper, Implications of Energy Peak for Collider Phenomenology: Top Quark Mass Determination and Beyond** Apr 2022
   K. Agashe, S. Airen, R. Franceschini, D. Kim, D. Sathyan
   arxiv.org/abs/2204.02928

1. **Snowmass2021 White Paper: Collider Physics Opportunities of Extended Warped Extra-Dimensional Models** Mar 2022
   K. Agashe, J.H. Collins, P. Du, M. Ekhterachian, S. Hong, D. Kim, R.K. Mishra, D. Sathyan
   arxiv.org/abs/2203.13305

Muon g-2 Articles

3. **The straw tracking detector for the Fermilab Muon g-2 Experiment** Nov 2021
   B.T. King et al.
   arxiv.org/abs/2111.02076 (JINST 17 (2022) P02035)

2. **Beam dynamics corrections to the Run-1 measurement of the muon anomalous magnetic moment at Fermilab** Apr 2021
   Muon g-2 Collaboration
   arxiv.org/abs/2104.03240 (Phys.Rev.Accel.Beams 24 (2021) 044002)

1. **Measurement of the Positive Muon Anomalous Magnetic Moment to 0.46 ppm** Apr 2021
   Muon g-2 Collaboration
   arxiv.org/abs/2104.03281 (Phys.Rev.Lett. 126 (2021) 141801)

Seminars

7. **CMS Exotica general meeting, May 2024**
   Unification of Searches and Measurements: Probing BSM with the W boson mass measurement

6. **Seminar at Texas A&M, December 2023**
   A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

5. **Seminar at UT Austin, December 2023**
   A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

4. **CMS SUSY general meeting, November 2023**
   A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

3. **HEP Seminar at University of Pittsburgh, November 2023**
   A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

2. **HEP Seminar at Northwestern University, October 2023**
   A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

1. **Theory Seminar at Washington University in St. Louis, October 2023**
   A New Purpose A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

Conference Talks

14. **Light Dark World 2026 at Carleton University, July 2026**
    Can the LHC be sensitive to the Light Dark World?

13. **CETUP* Workshop, June 2026**
    Producing the GeV Galactic Center Excess via Cosmic Ray-Dark Matter Scattering

12. **Particle Physics on the Plains at the University of Kansas, November 2025**
    A Baryon and Lepton Number Violaton Model Testable at the LHC

11. **CETUP* Workshop, June 2025**
    Can the LHC be sensitive to light dark mediators?

10. **Phenomenology Conference at the University of Pittsburgh, May 2025**
    Can the LHC be sensitive to light dark mediators?

9. **Particle Physics on the Plains at the University of Kansas, November 2024**
   A comprehensive analysis of supernova neutrino-dark matter interactions

8. **Texas TACOS @ UT Austin, October 2024**
   A comprehensive analysis of supernova neutrino-dark matter interactions

7. **NuFact Conference, September 2024**
   A comprehensive analysis of supernova neutrino-dark matter interactions

6. **Mitchell Conference at Texas A&M University, May 2024**
   A comprehensive analysis of supernova neutrino-dark matter interactions

5. **Particle Physics on the Plains at the University of Kansas, October 2023**
   A New Purpose for the W-mass Measurement: Searching for New Physics via l + MET

4. **Phenomenology Conference at the University of Pittsburgh, May 2023**
   Probing Dark Matter-Neutrino Interactions via Supernova Neutrinos

3. **Phenomenology Conference at the University of Pittsburgh, May 2022**
   Model-Independent Measurement of Top Quark Mass Using B-Hadron Decay Lengths (Part I)

2. **April APS Meeting, April 2021**
   Signals of KK graviton from extended warped extra dimensions at the LHC (II)

1. **Phenomenology Conference at the University of Pittsburgh, May 2020**
   Signals of KK graviton from extended warped extra dimensions at the LHC (II)

Conferences Organized

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2026
Texas A&M University, May 27-30, 2026

The Mitchell Conference on Collider, Dark Matter, and Neutrino Physics 2025
Texas A&M University, May 13-16, 2025

Awards

Breakthrough Prize in Fundamental Physics Apr 2026
Muon g-2 Collaboration

Schools Attended

GGI Lectures on the Theory of Fundamental Interactions Jan 2023 – Jan 2023

TASI 2024: The Frontiers of Particle Theory June 2024 – June 2024

Outreach

BU Student Mentorship Program PRISM (2016-2018)
Mentored incoming freshman undergraduates in physics, providing guidance on courses, research opportunities, and resources

TAMU Physics and Engineering Festival (2025-2027)
Presented demonstrations of fluid dyanmics to the general public, including high school students and families

University of South Dakota talk: "My path through particle physics research so far" (December 2024)
Presented my path through particle physics research to students, showing a variety of interesting ideas to pursue

Skills

Programming:
Mathematica, Python, C++

Physics Software:
MadGraph5_aMC@NLO, Pythia6, Pythia8, Delphes, ROOT, GEANT4, FeynRules